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Principles of Macroeconomics

Lecture 4: Unemployment and the Labour Market

Prof. Dr. Jan-Egbert Sturm

Exam Information

- Exam dates:
  - Final exam:
    - **Tuesday, 13.01.2026, 14:00-15:30** (Rooms to be announced)
  - Retake/repetition exam (only for students that failed in the final exam)
    - Monday, 23.02.2026, 14:00-15:30 (Rooms to be announced)
- Exam registration
  - Please register for the exam on myStudies
    - Enrolment in the course is NOT equal to an exam registration.
  - Further information:
    - https://ethz.ch/applications/teaching/en/applications/mystudies/matriculation/examinations.html
    - https://ethz.ch/students/en/studies/performance-assessments/registration.html
- Type of exam:
  - 90 minutes written paper exam
  - Closed book

Schedule

Date	Lecture	Topic	Chapters
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Lecture 3: Production and Growth

Prof. Dr. Jan-Egbert Sturm

How Productivity Is Determined

- ***Physical capital*** per worker is the stock of equipment and structures that are used to produce goods and services
  - It includes machines and tools build or repair automobiles or furniture / Office buildings, schools, etc.
  - Physical capital is a produced factor of production
    - It is an input into the production process that in the past was an output from that process
- ***Human capital*** per worker is the economist's term for the knowledge and skills that workers acquire through education, training, and experience
  - Like physical capital, human capital raises a nation's ability to produce goods and services
- ***Natural resources*** are inputs used in production that are provided by nature
  - Renewable resources include trees and forests
  - Nonrenewable resources include petroleum and coal
    - Natural resources can be important but are not necessary for an economy to be highly productive
- ***Technological knowledge*** includes society's understanding of the best ways to produce
  - Human capital includes the resources expended transmitting this understanding to the labour force

FYI: The Production Function

- Economists often use a production function to describe the relationship between the quantity of inputs used in production and the quantity of output from production
- $Y = A\,F(L, K, H, N)$
  - Y = quantity of output
  - A = available production technology
  - L = quantity of labour
  - K = quantity of physical capital
  - H = quantity of human capital
  - N = quantity of natural resources
  - $F(\ )$ is a function that shows how the inputs are combined
- A production function has constant returns to scale if, for any positive number x,
  - $xY = A\,F(xL, xK, xH, xN)$
- A doubling of all inputs causes the amount of output to double as well
- Production functions with constant returns to scale have an interesting implication
- Setting $x = 1/L$ results in:
  - $Y/L = A\,F(1, K/L, H/L, N/L)$
  - Where:
    - Y/L = output per worker = labour productivity
    - K/L = physical capital per worker
    - H/L = human capital per worker
    - N/L = natural resources per worker

The Determinates Of Economic Growth

- The Solow growth model helps us understand some key elements of determinants of economic growth
  - GDP in any country can be assumed to be an extension of a firm's production function where the level of output is dependent on the factors of production employed
- Assumptions are
  - Two production factors: labour (L) and capital (K) (next to technology: A)
    - Output (GDP) = Y = AF(K,L)
  - Constant returns to scale: xY = AF(xK,xL)
    - This implies that capital and labour are each subject to ***diminishing marginal returns***
  - A closed economy and no government (G=0 & NX=0)
    - Therefore, Y = C + I + G + NX simplifies to Y = C + I

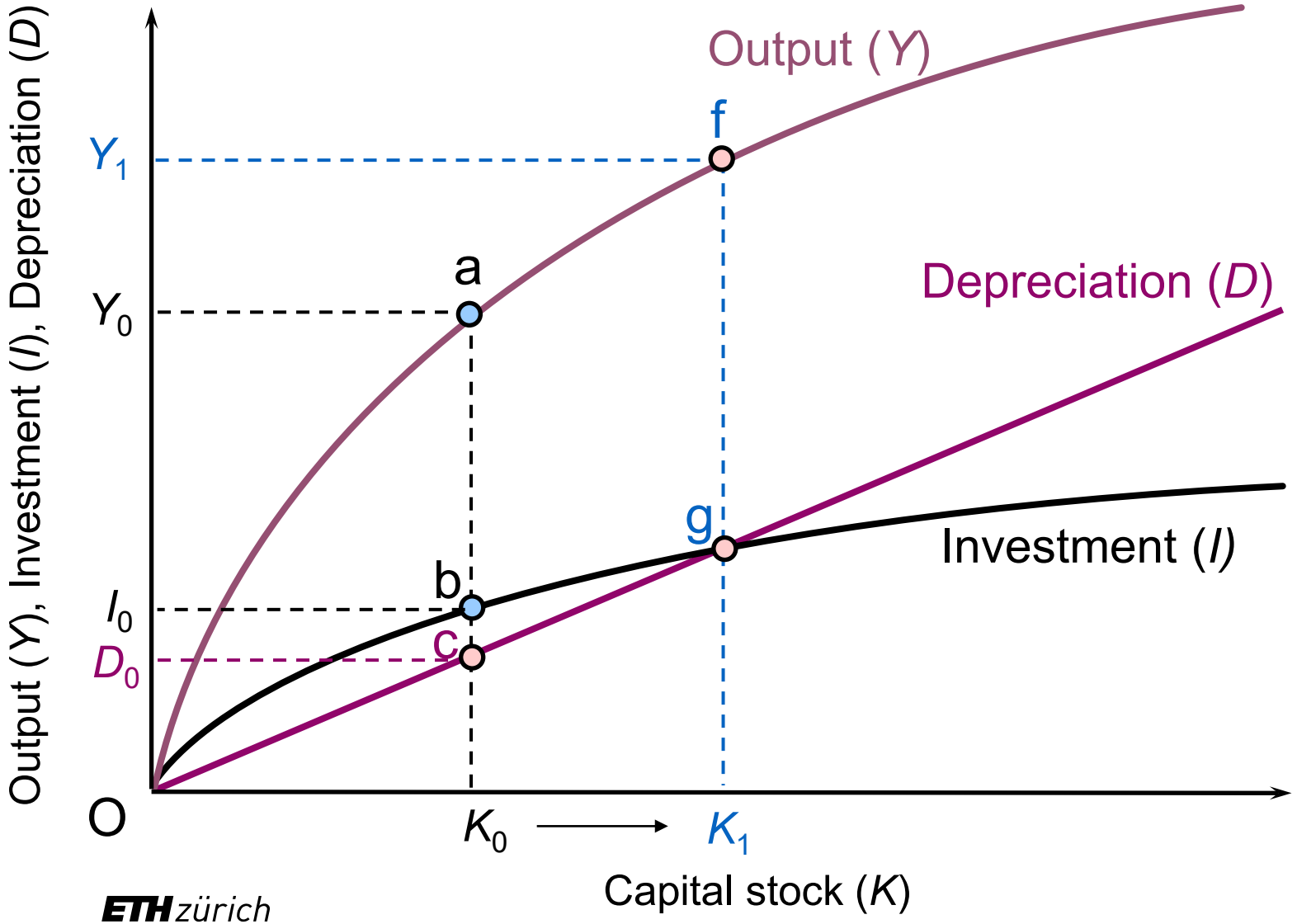
The Meaning of Saving and Investment

- For the economy as a whole, saving must be equal to investment:

$$(Y - C) \equiv S = I$$

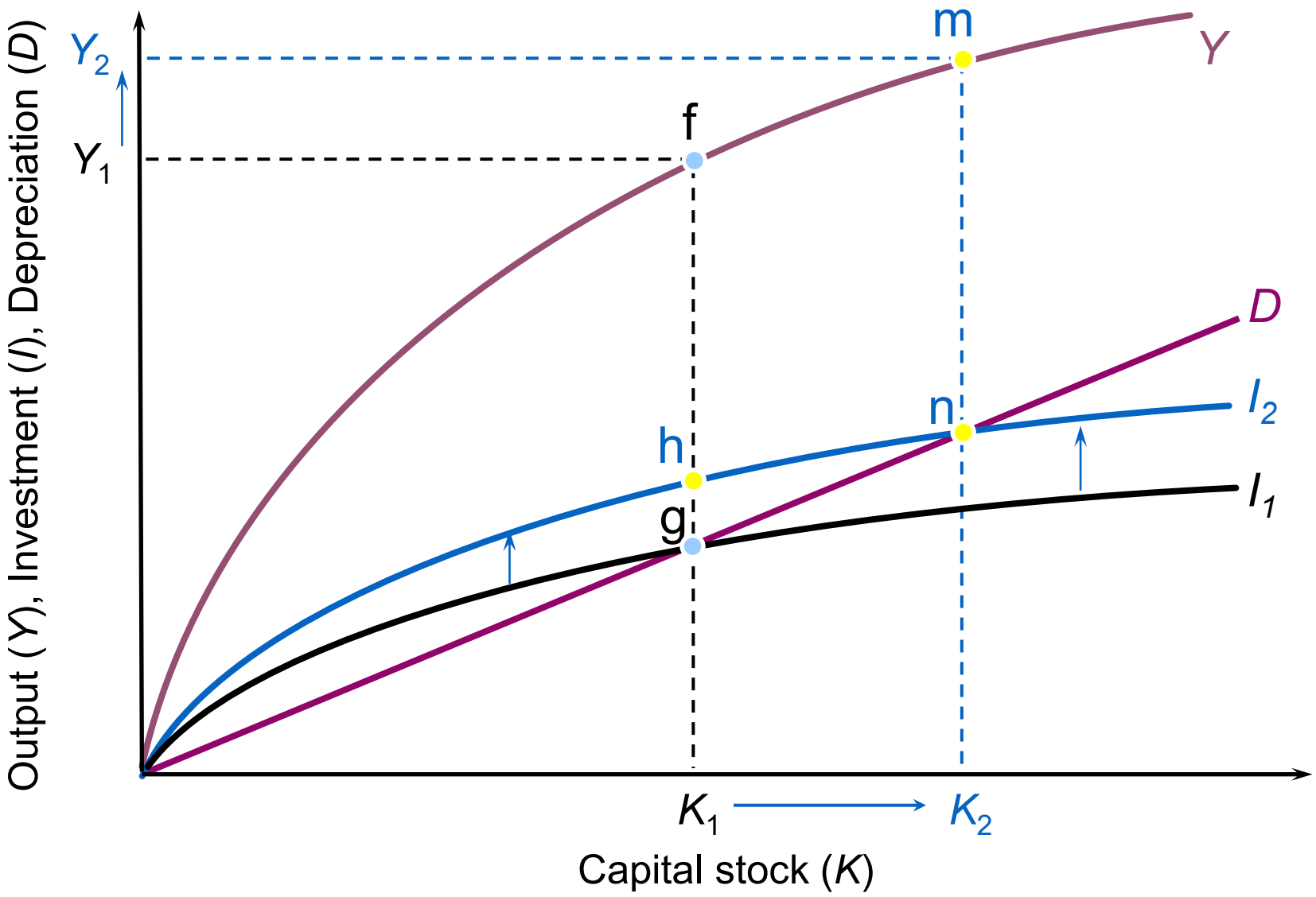
  - Note: “Investment” means the creation of new productive assets by firms and households, e.g. building a ***new*** factory or a ***new*** house
  - Buying an old house or a share in an existing company is only a financial transaction, although this is also often called “investment”
- Assume a constant share of income is saved and therefore invested: $S = sY$
- Firms invest in order to raise future productivity, or to replace depreciated capital
  - Assumption: capital depreciates with a constant rate: $D = \delta K$
- The difference between total investment (I) and depreciation (D) is called net investment

Investment, Growth, and Output

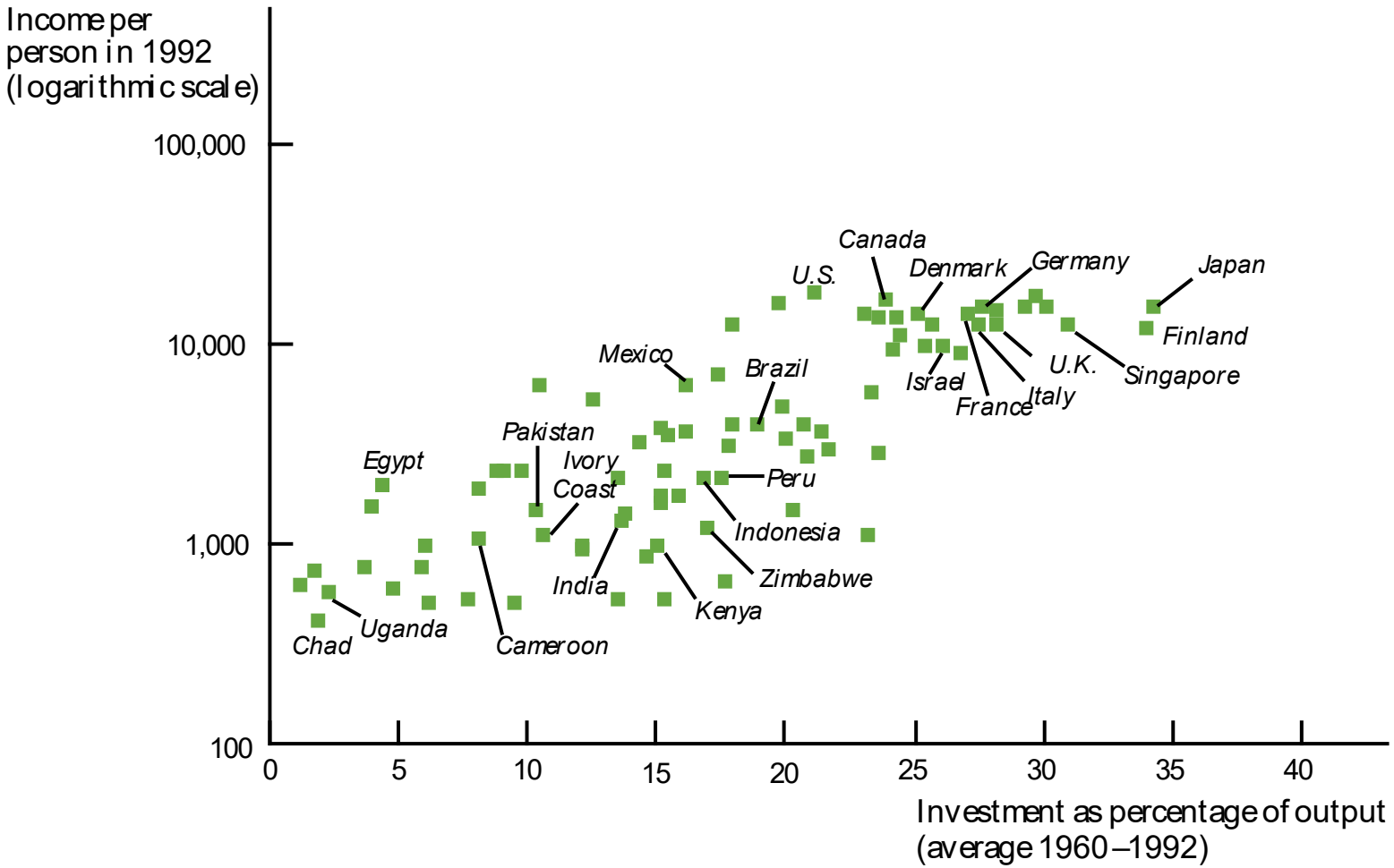


- For given A and L, increasing K is associated with rising GDP
- Diminishing marginal product of K implies with higher K lower GDP growth
- The distance between GDP and the investment level is consumption
- At a given saving rate, with positive net investment, the capital stock (and hence output) will grow until *D*=*I*
- ***Steady-state equilibrium*** is the point in a growing economy where investment is the same as spending on depreciation and the capital-output ratio remains constant
- Increasing the saving rate, allows K and Y to grow further, again until *D*=*I*

Effect of increase in the savings rate



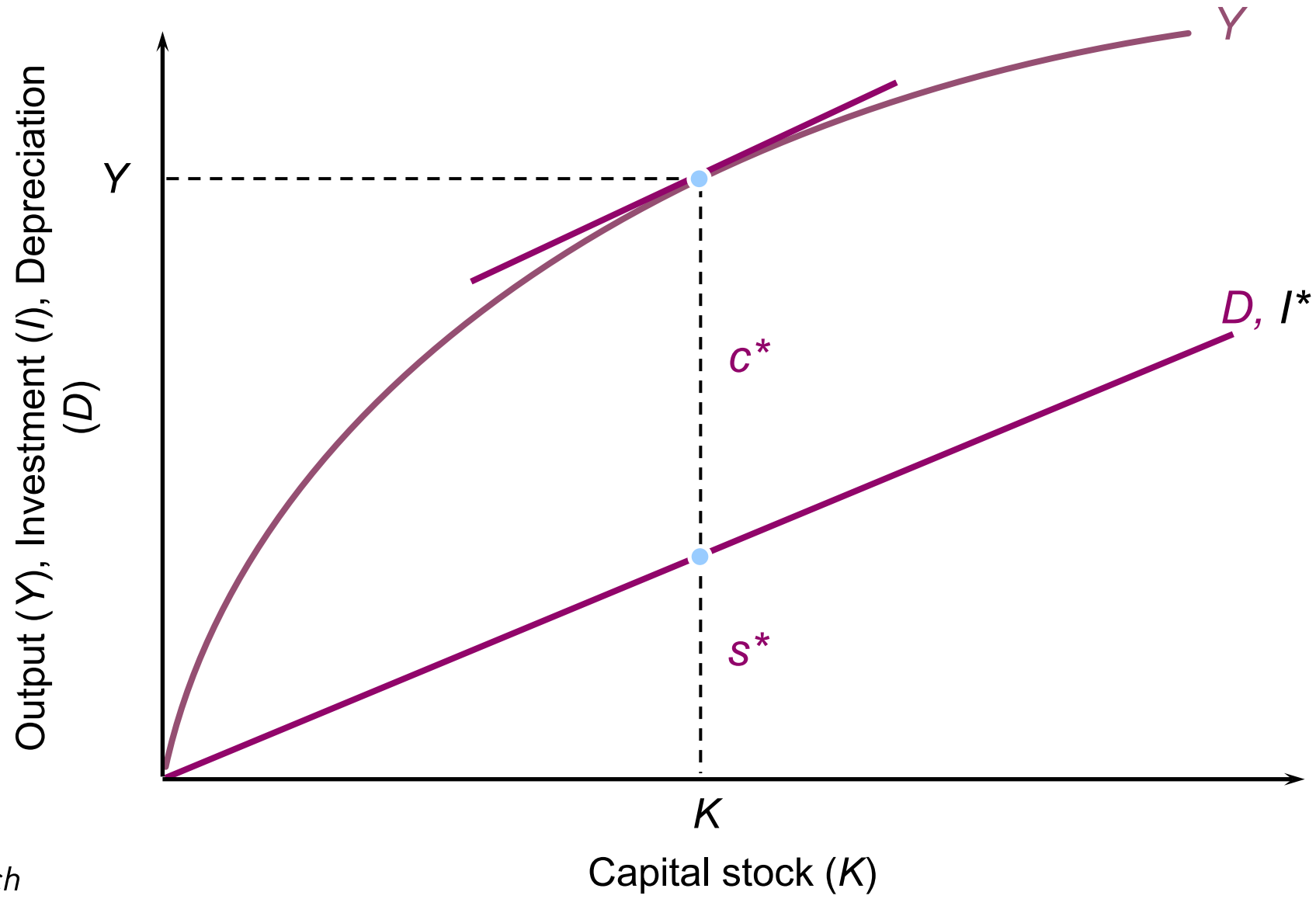
International Evidence on Investment Rates and Income per Person



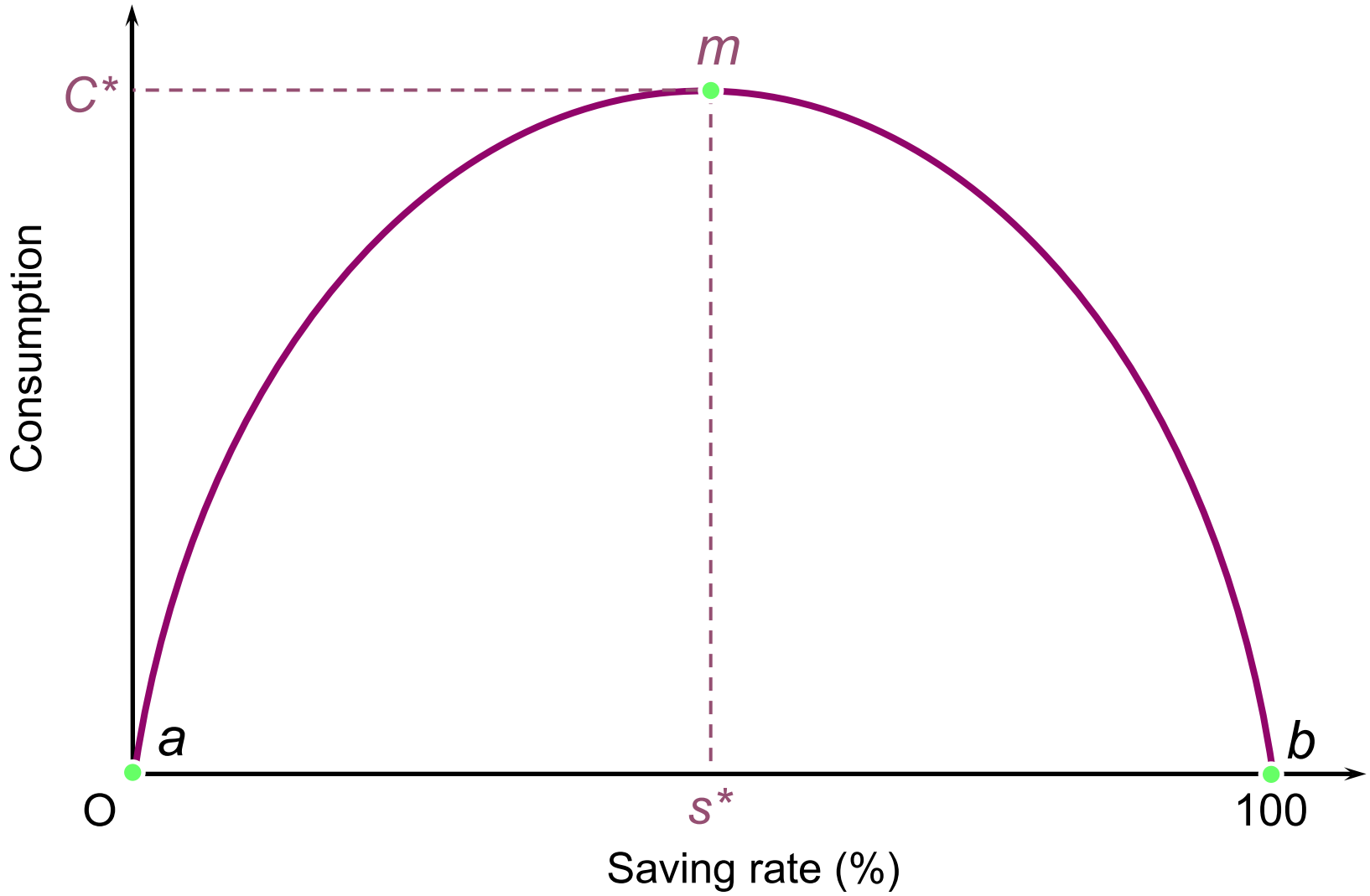
Investment, Growth and Output

- The economy's output is used for saving (investment) or consumption
- Long-run (steady state) consumption is the difference between steady state output and steady state depreciation (investment)
- There's a unique saving rate that maximizes the difference between output and investment

Steady state consumption



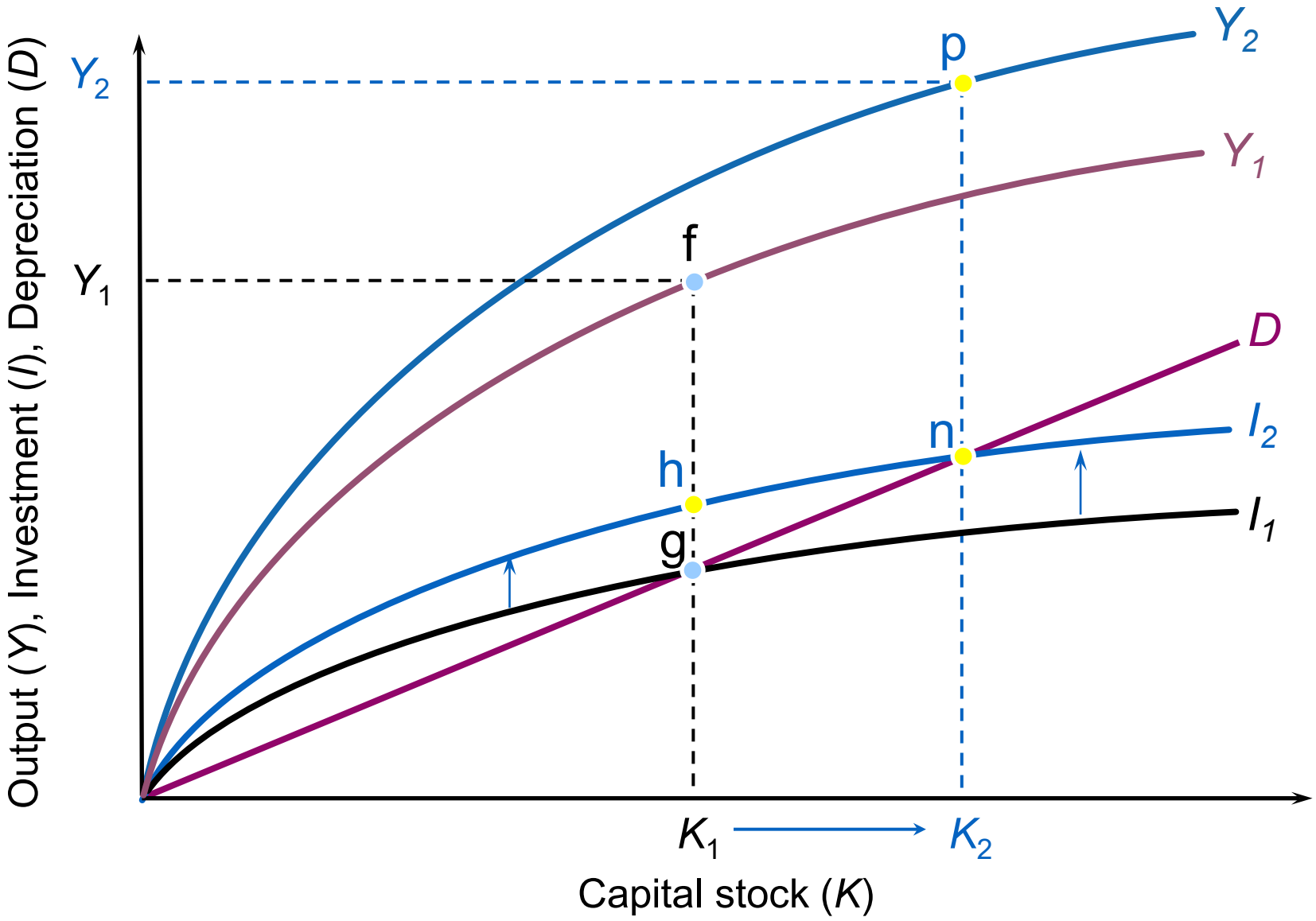
An optimum saving rate



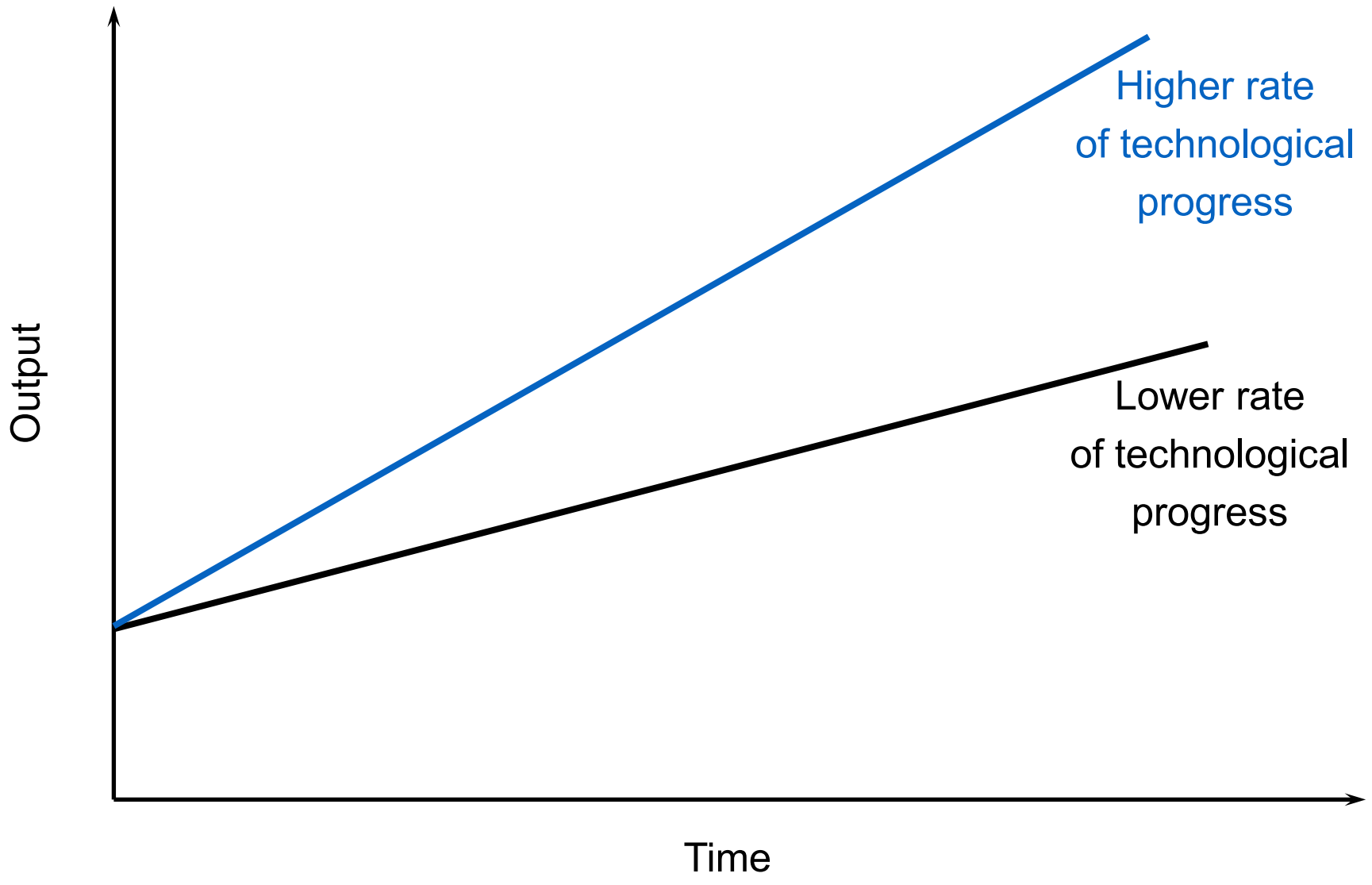
Technological Progress

- So far, the growth rate in the steady state is zero (A is constant)
- This is no longer the case if we allow for technological progress

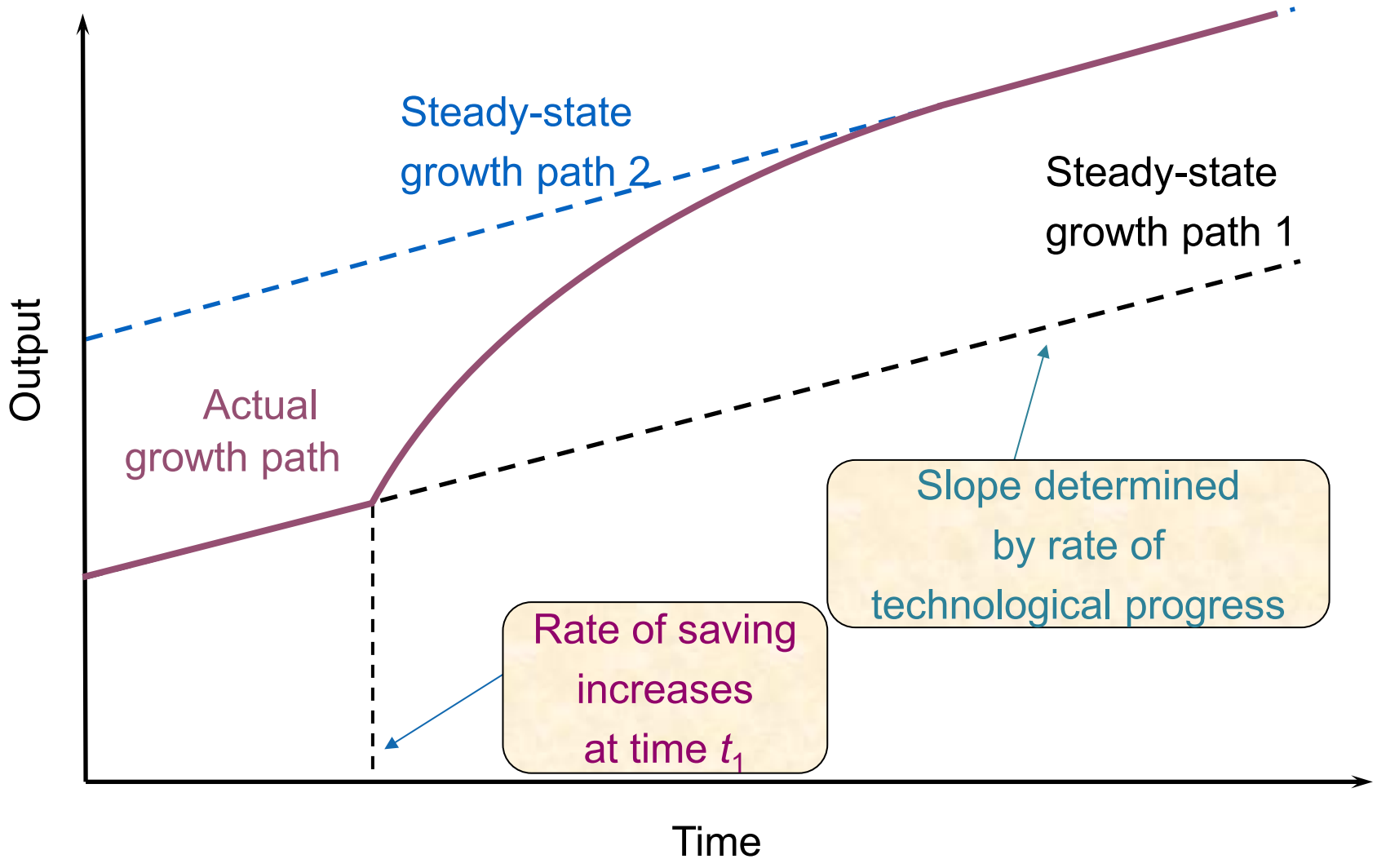
Effect of technological progress



Effect of technological progress



Effect of an increase in the saving rate with given technological progress



Long-run Equilibrium

- For an economy with a given level of physical capital, the growth path will be dependent on the:
  - Level of technology
  - Productivity of labour and capital
  - Savings rate which determines investment in physical capital
- The ratio of capital to labour is one aspect of how productive labour can be
- Countries like Korea and Vietnam have a supply of relatively cheap labour and have invested in capital and as a result labour productivity is relatively high and there is strong economic growth
- In many African countries investment in capital stock is extremely low and so labour productivity is also very low and there is weak economic growth
- The Solow model provides a means of understanding the transition of economies over time
  - Positive net investment in capital will increase capital per worker and lead to growth
  - However, investment is determined by the savings ratio
    - In less developed countries this may be relatively low because incomes are low
  - Also it is difficult to do business in these countries and therefore do not reach the steady state

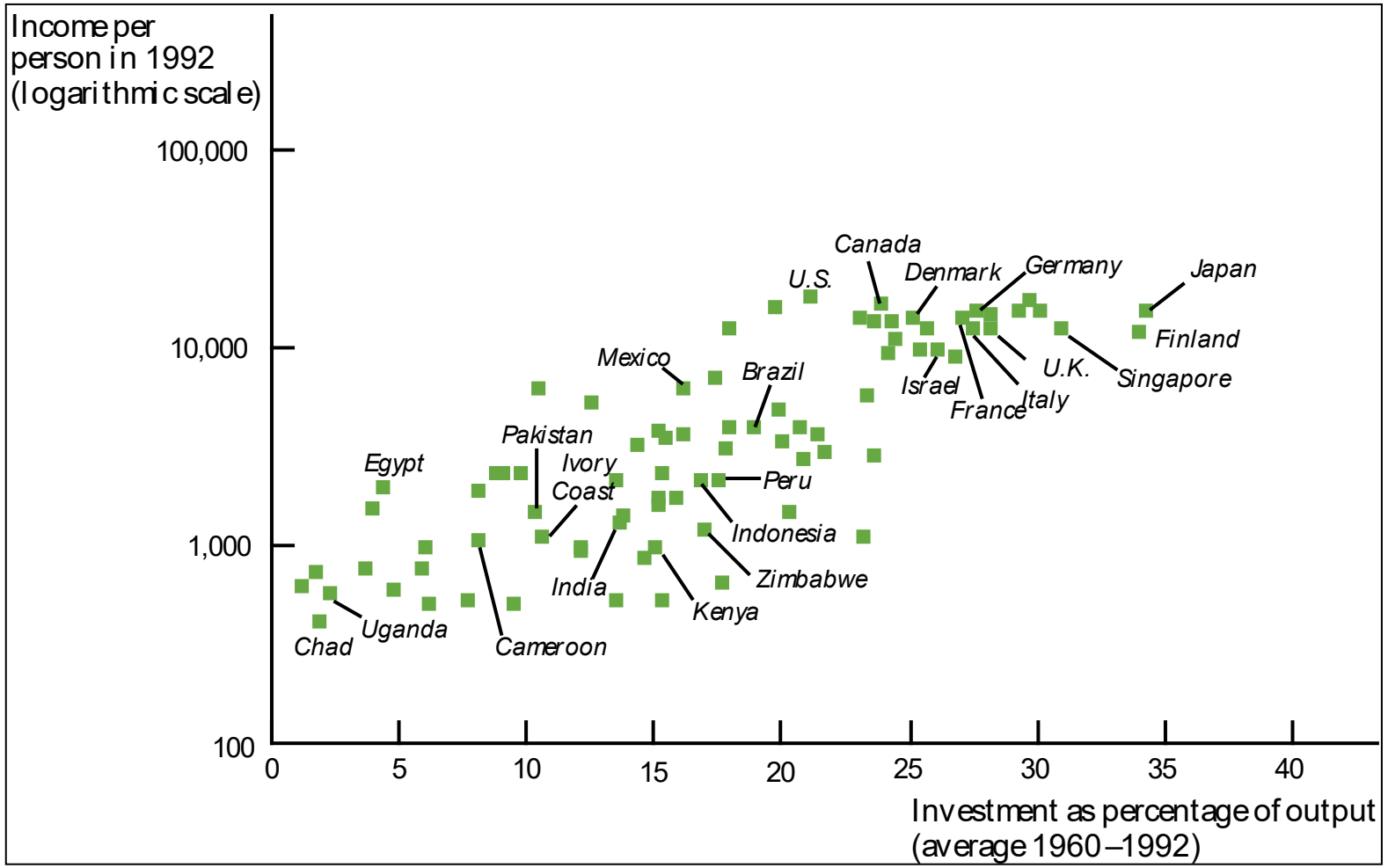
ECONOMIC GROWTH AND PUBLIC POLICY

- Government policies that raise productivity and living standards
  - Encourage saving and investment
  - Encourage investment from abroad
  - Encourage education and training
  - Establish secure property rights and maintain political stability
  - Promote free trade
  - Promote research and development

Saving and Investment

- One way to raise future productivity is to invest more current resources in the production of capital
- Since resources are scarce, producing more capital requires producing fewer consumption goods
- Reducing consumption = increasing saving
  This extra saving funds the production of investment goods
- Hence, a trade-off between current and future consumption

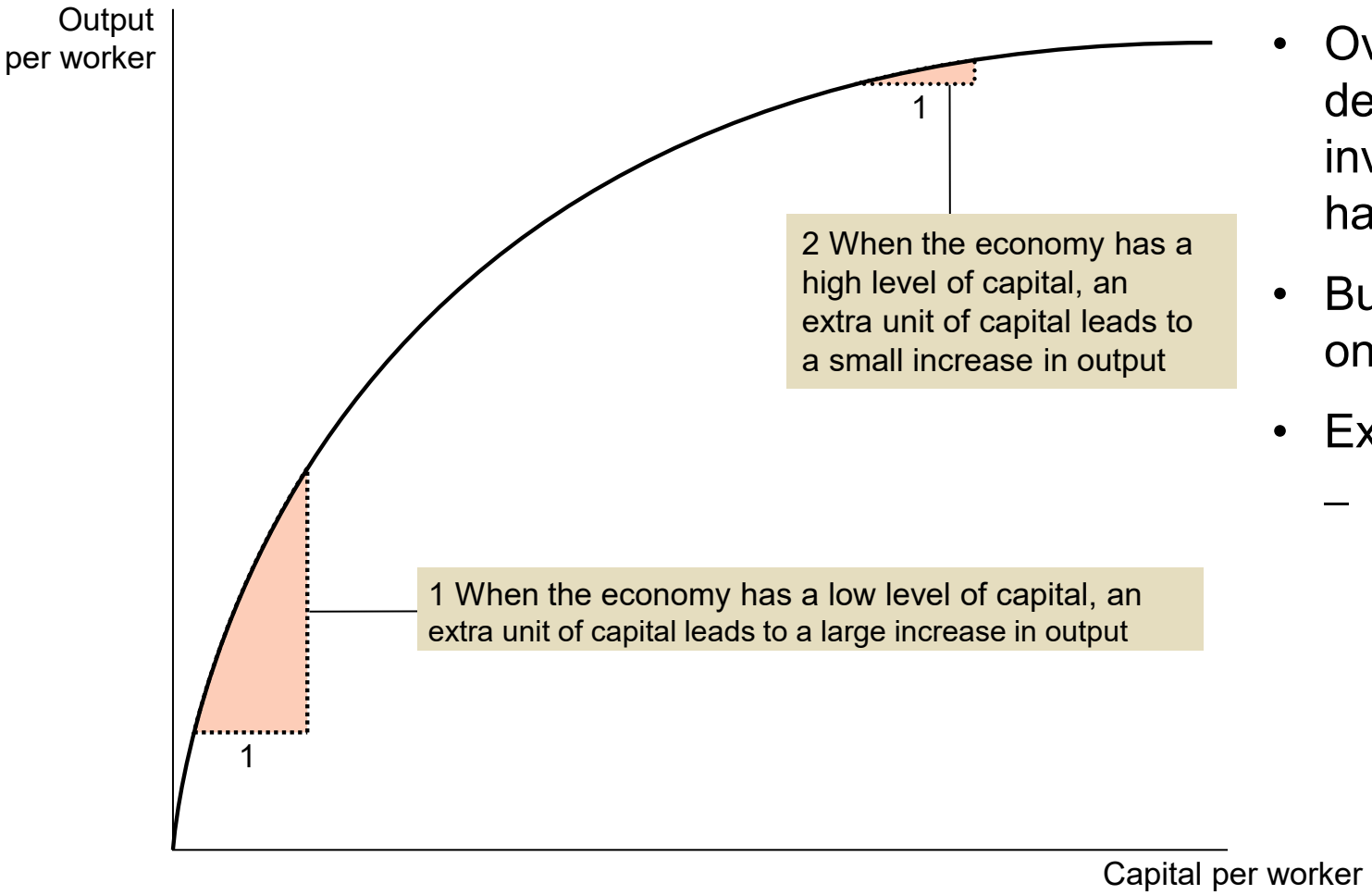
International Evidence on Investment Rates and Income per Person



Diminishing Returns and the Catch-Up Effect

- As the stock of capital rises, the extra output produced from an additional unit of capital falls; this property is called ***diminishing returns***
- Because of diminishing returns, an increase in the saving rate leads to higher growth only for a while
- In the long run, the higher saving rate leads to a higher level of productivity and income, but not to higher growth in these areas
- The ***catch-up effect*** refers to the property whereby countries that start off poor tend to grow more rapidly than countries that start off rich

Illustrating the Production Function and an example of the Catch-Up Effect



- Over 1960-1990, the US and South Korea devoted a similar share of GDP to investment, so you might expect they would have similar growth performance
- But growth was >6% in South Korea and only 2% in the US
- Explanation: the catch-up effect
  - In 1960, **K/L** was far smaller in South Korea than in the US, hence South Korea grew faster

Investment from Abroad

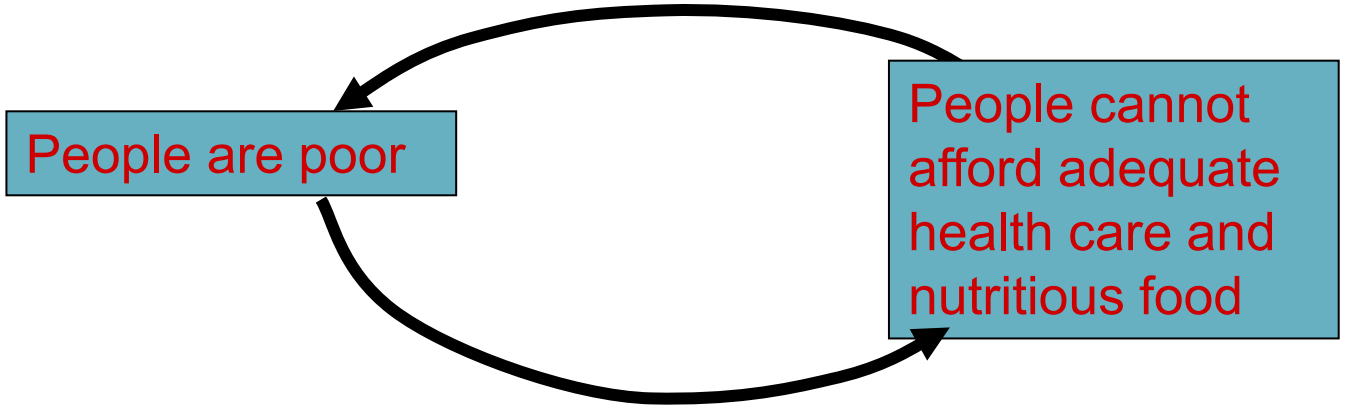
- Governments can increase capital accumulation and long-term economic growth by encouraging investment from foreign sources
- Investment from abroad takes several forms:
  - Foreign Direct Investment
    - Capital investment owned and operated by a foreign entity
  - Foreign Portfolio Investment
    - Investments financed with foreign money but operated by domestic residents
- Some of the returns from these investments flow back to the foreign countries that supplied the funds
- Especially beneficial in poor countries that cannot generate enough saving to fund investment projects themselves
- Also helps poor countries learn state-of-the-art technologies developed in other countries

Education

- For long-run growth, education is at least as important as investment in physical capital
  - Investing in human capital also involves a trade-off between the present and the future:
    - Spending a year in school requires sacrificing a year's wages now to have higher wages later
  - In the US, each year of schooling raises a person's wage, on average, by about 10 percent
- Government can enhance the standard of living by providing schools and encourage the population to take advantage of them
  - Positive externality: An educated person might generate new ideas about how best to produce goods and services, which in turn, might enter society's pool of knowledge and provide an external benefit to others
- One problem facing some poor countries is the brain drain
  - the emigration of many of the most highly educated workers to rich countries

Health and Nutrition

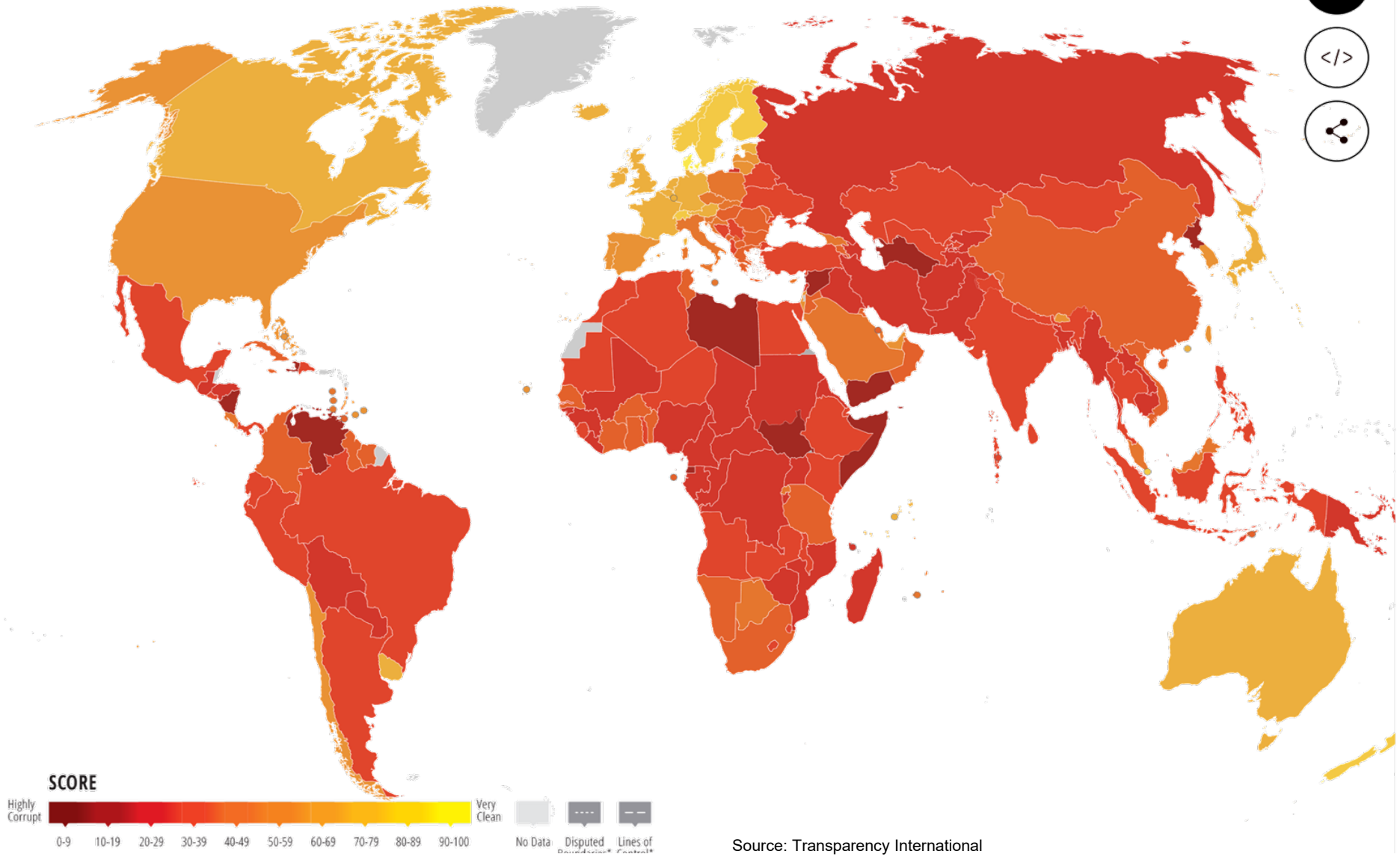
- Healthier workers are more productive
  - Health care expenditure is a type of investment in human capital
    - healthier workers are more productive
- Investments in the health of the population can lead to increase living standards
- Countries can get caught in a vicious cycle



Property Rights and Political Stability

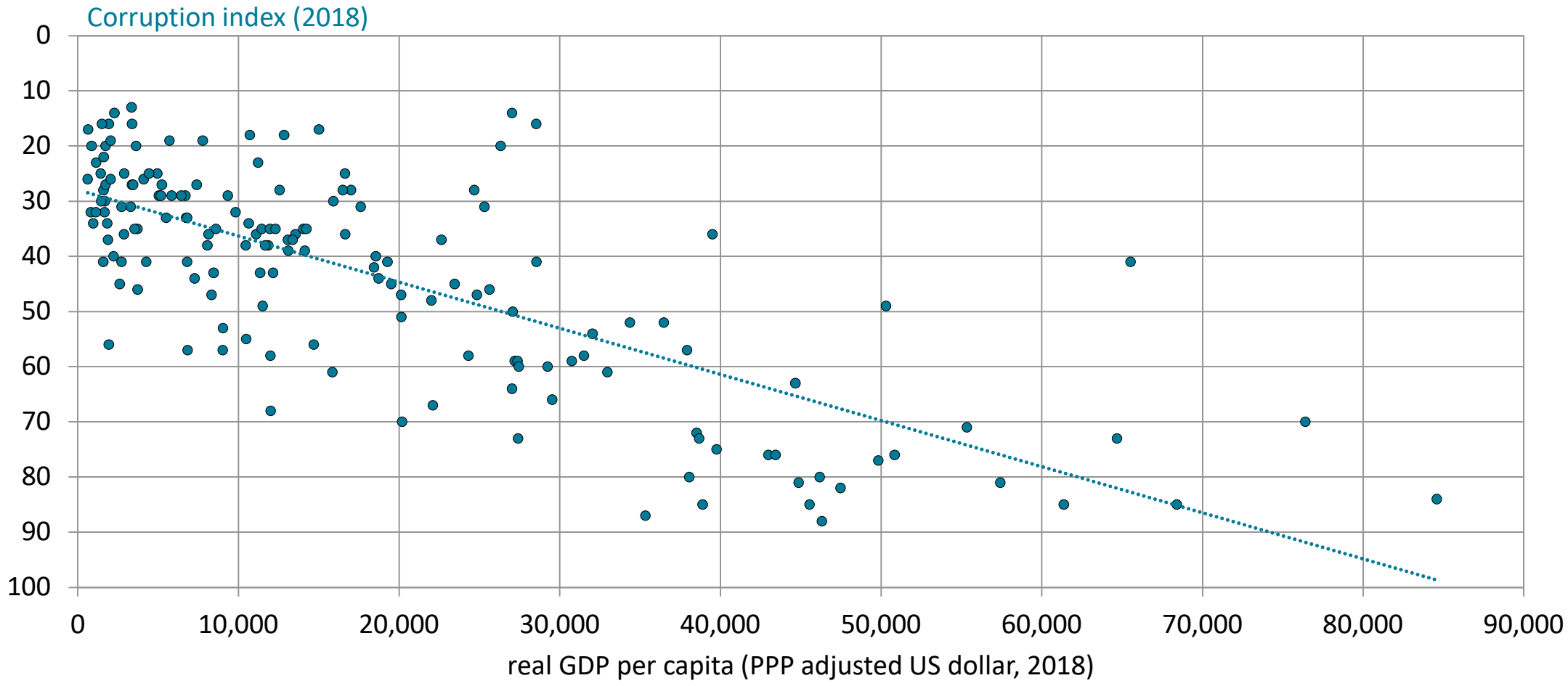
- Recall: Markets are usually a good way to organize economic activity
  - The price system allocates resources to their most efficient uses
- This requires respect for property rights
- Property rights refer to the ability to exercise authority over resources one owns
- An economy-wide respect for property rights is an important prerequisite for the price system to work
- In many poor countries, the justice system doesn't work very well:
  - Contracts aren't always enforced
  - Fraud, corruption often go unpunished
  - In some, firms must bribe government officials for permits
- It is necessary for investors to feel that their investments are secure
  - Political instability (e.g., frequent coups) creates uncertainty over whether property rights will be protected in the future

Corruption around the World (2023 Index)



Source: Transparency International

Real GDP per capita and corruption around the World (2018)



Sources: Transparency International, GGDC

Free Trade

- Trade is, in some ways, a type of technology
- A country that eliminates trade restrictions will experience the same kind of economic growth that would occur after a major technological advance
- Some countries engage in …
  - … inward-orientated trade policies, avoiding interaction with other countries
  - … outward-orientated trade policies, encouraging interaction with other countries

- Countries and regions with inward-oriented policies have generally failed to create growth
  - E.g., Argentina during the 20th century
- Countries and regions with outward-oriented policies have often succeeded
  - E.g., South Korea, Singapore, Taiwan after 1960

Technological progress - Research and Development

- The advance of technological knowledge has led to higher standards of living
- One reason is that knowledge is a public good:
  - Ideas can be shared freely, increasing the productivity of many
- Technology can offset the effects of diminishing marginal product
- Most technological advances come from private research by firms & individuals
  - Innovation and R&D are important ways in which technology growth can arise and can provide competitive advantage for firms over rivals which are both distinctive and defensible
- Government can encourage the development of new technologies through
  - Patent laws
  - Tax incentives or direct support for private sector R&D
  - Grants for basic research at universities

Population Growth

- Population growth interacts with other factors of production:
  - Stretching natural resources
    - 200 years ago, Malthus argued that population growth meant that the world was doomed to live in poverty forever
    - Since then, the world population has increased sixfold. If Malthus was right, living standards would have fallen. Instead, they've risen
    - He failed to understand that new ideas would be developed to increase the production of food and other goods, fertilizers, mechanized equipment, and new crop varieties (technological progress)
  - Diluting the capital stock
    - Lower K/L = lower productivity & living standards
    - more children = greater strain on educational system = lower H/L
  - Promoting technological progress
    - more scientists, inventors, engineers = more frequent discoveries
      - growth rates increased as the world's population increased
      - more populated regions grew faster than less populated ones

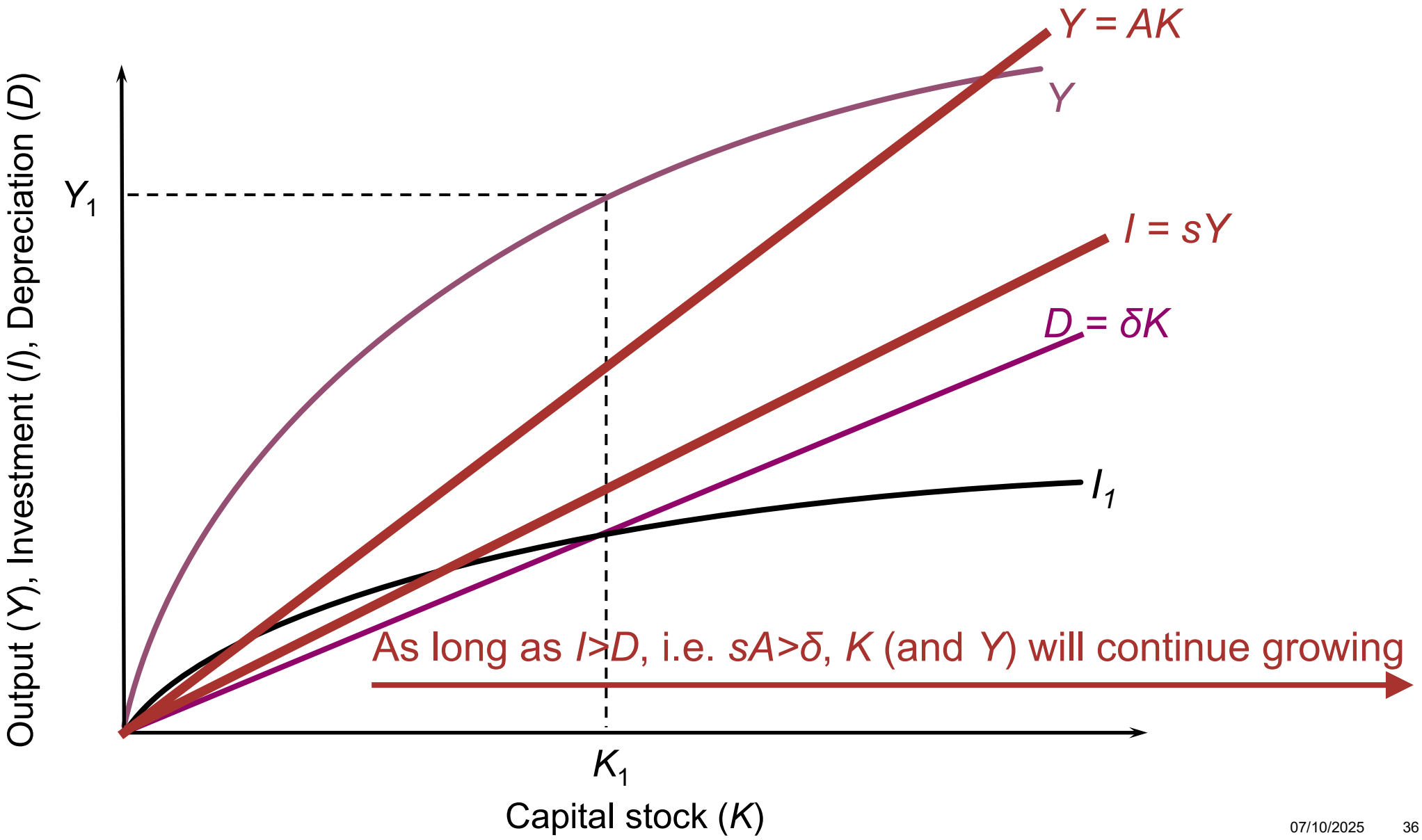
Are Natural Resources a Limit to Growth?

- Some argue that population growth is depleting the Earth's non-renewable resources, and thus will limit growth in living standards
- But technological progress often yields ways to avoid these limits:
  - Hybrid cars use less gas
  - Better insulation in homes reduces the energy required to heat or cool them
- As a resource becomes scarcer, its market price rises, which increases the incentive to conserve it and develop alternatives

Endogenous Growth Theory

- ***Endogenous growth theory*** is a theory of long-run economic growth which results from the creation of new knowledge and technology which impacts on everyone and makes them more productive as a result
- A basic endogenous growth model:
  - Production function: $Y = A K$
    where A is the amount of output for each unit of capital (A is exogenous & constant)
  - Key difference between this model & Solow:
  - Marginal product of capital is constant here, while it diminishes in the Solow model
- As in the Solow model: Investment = sY and Depreciation = δK
  - Equation of motion for total capital: $\Delta K = sY - \delta K$ or $\Delta K/K = sA - \delta$
  - If $sA > \delta$, then income will grow forever, and investment is the "engine of growth"
  - Here, the permanent growth rate depends on s – in the Solow model, it does not

A basic endogenous growth model



Summary

- Economic prosperity, as measured by real GDP per person, varies substantially around the world
- The average income of the world's richest countries is more than ten times that in the world's poorest countries
- The standard of living in an economy depends on the economy's ability to produce goods and services
- Productivity depends on the amounts of physical capital, human capital, natural resources, and technological knowledge available to workers
- Government policies can influence the economy's growth rate in many different ways
- The Solow model notes that the accumulation of capital is subject to diminishing returns
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Summary

- Productivity depends on the amounts of physical capital, human capital, natural resources, and technological knowledge available to workers
- Government policies can influence the economy's growth rate in many different ways
- The Solow model notes that the accumulation of capital is subject to diminishing returns
- Because of diminishing returns, higher saving leads to a higher growth for a period of time, but growth will eventually slow down
- Also because of diminishing returns, the return to capital is especially high in poor countries
- Endogenous growth theory explains how technology can change and offset the effects of diminishing marginal productivity. Innovation and R&D are important ways in which technology growth can arise
- Government policies can try to influence the economy's growth rate in many ways: by encouraging saving and investment, encouraging investment from abroad, fostering education, maintaining property rights and political stability, allowing free trade, promoting the research and development of new technologies, and controlling population growth

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Principles of Macroeconomics

Lecture 4: Unemployment and the Labour Market

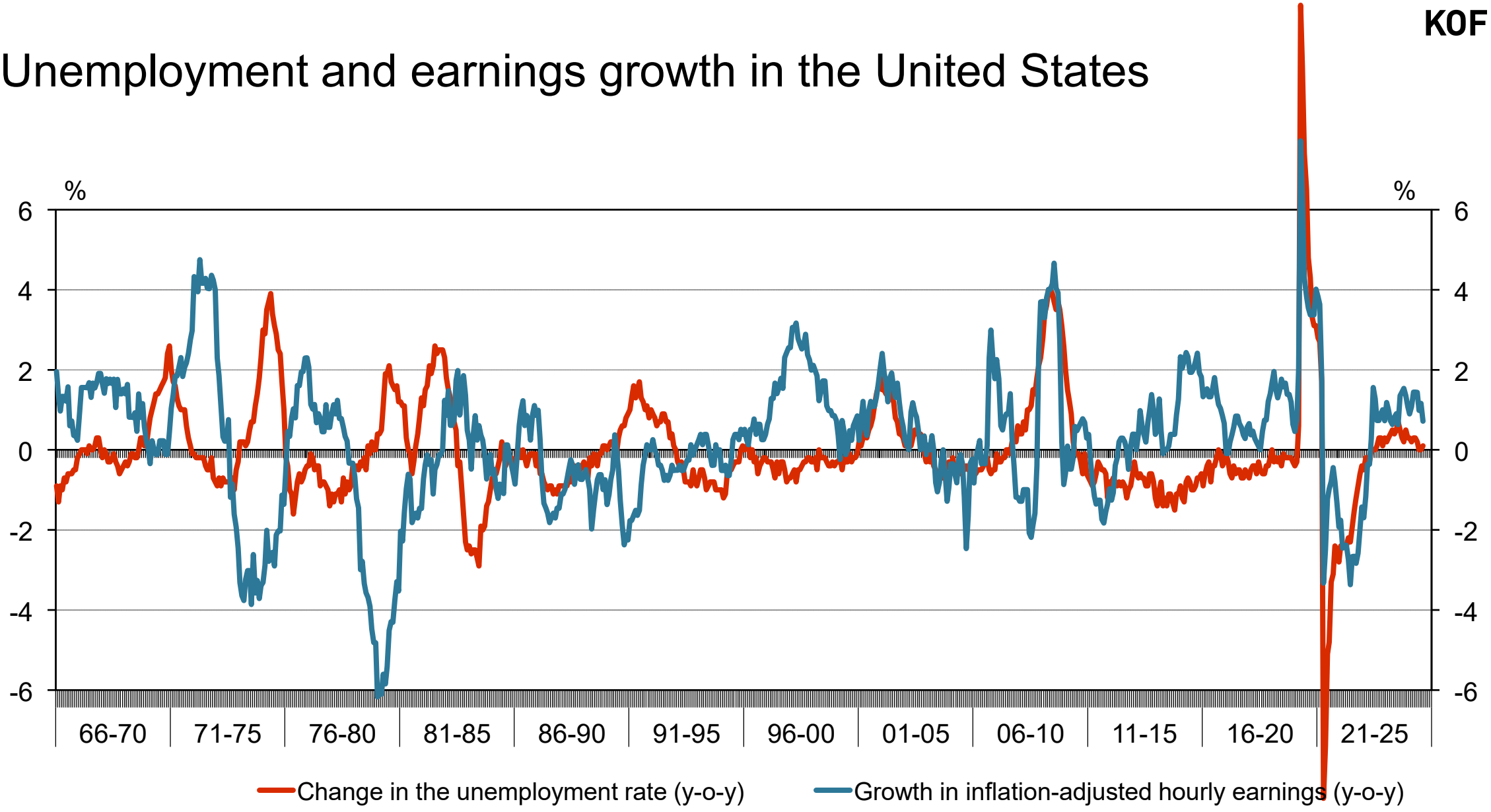
Prof. Dr. Jan-Egbert Sturm

Unemployment: Social Problems, Tensions and Costs

- Unemployment incurs costs to the individual (loss of earnings, stress, self-esteem and health problems, family breakdown, de-skilling etc.) as well as costs to the society and economy (opportunity costs, tax and benefits effects, etc.)

- Empirical studies for the U.S. show that each percentage-point increase in the unemployment rate is associated with:
  - 920 more suicides
  - 650 more homicides
  - 4,000 more people admitted to state mental institutions
  - 3,300 more people sent to state prisons
  - 37,000 more deaths
  - increases in domestic violence and homelessness

Unemployment and earnings growth in the United States



Source: BLS

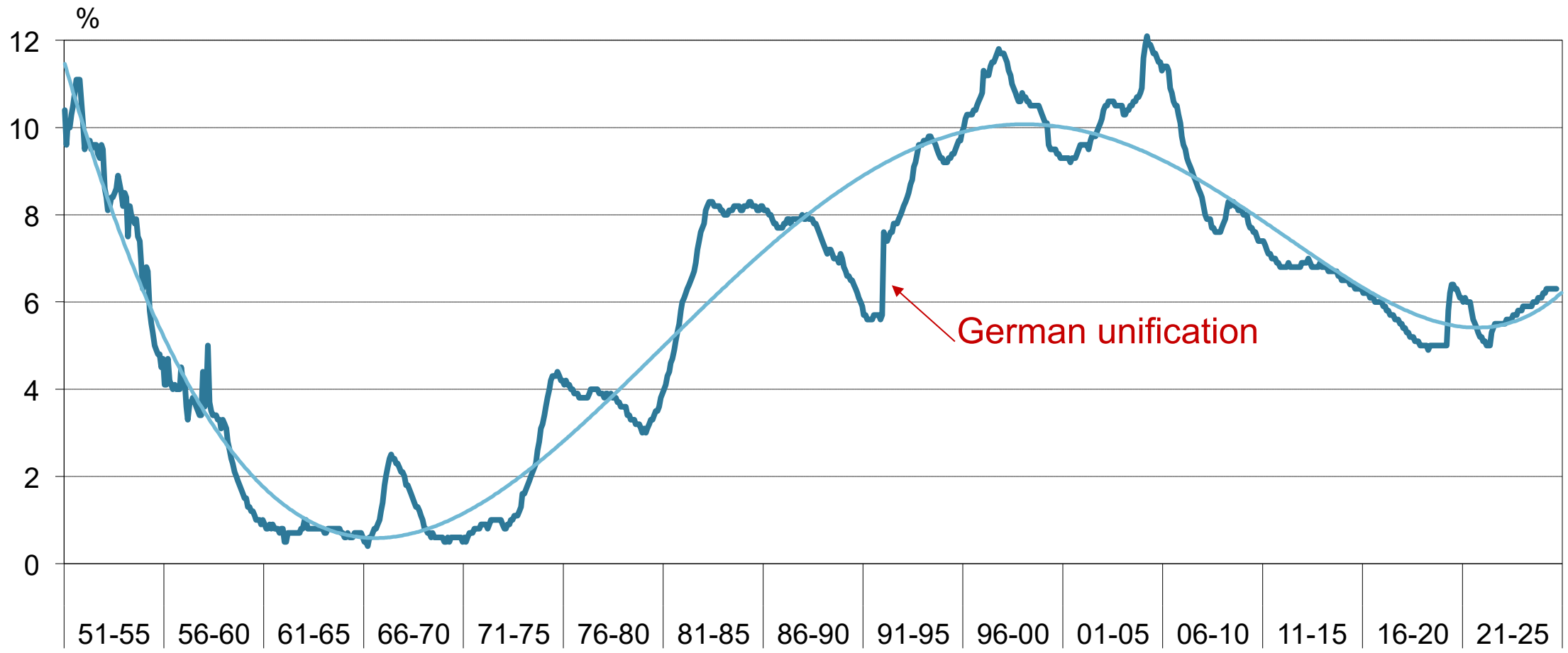
Categories of Unemployment

- The problem of unemployment is usually divided into two categories, the long-run problem and the short-run problem

1. Long-run Problem: ***Natural Rate of Unemployment***
   - The natural rate of unemployment is unemployment that does not go away on its own even in the long run
   - It is the amount of unemployment that the economy normally experiences

2. Short-run Problem: ***Cyclical Unemployment***
   - Cyclical unemployment refers to the year-to-year fluctuations in unemployment around its natural rate
   - It is associated with short-term ups and downs of the business cycle

Unemployment rate in Germany



Source: Deutsche Bundesbank

Describing Unemployment: Four Basic Questions

1. How does government measure the economy's rate of unemployment?
2. What problems arise in interpreting the unemployment data?
3. How long are the unemployed typically without work?
4. Why are there always some people unemployed?

(1) How Is Unemployment Measured? Definitions

- To measure unemployment national authorities aim at splitting the entire population into three groups:
  1. ***Employed***
     - A person with a job is considered as employed.
  2. ***Unemployed***
     - A person is considered as unemployed, if that person does not have a job and(!) is able and available (willing) to work at current wage rates.
  3. ***Not in the Labour Force***
     - People who do not have a job and are either not able or not available to work

- The ***Labour Force*** is the total number of (employed and unemployed) workers:
  Labour Force = Employed + Unemployed

Illustration: The Breakdown of Swiss Population at the start of 2023

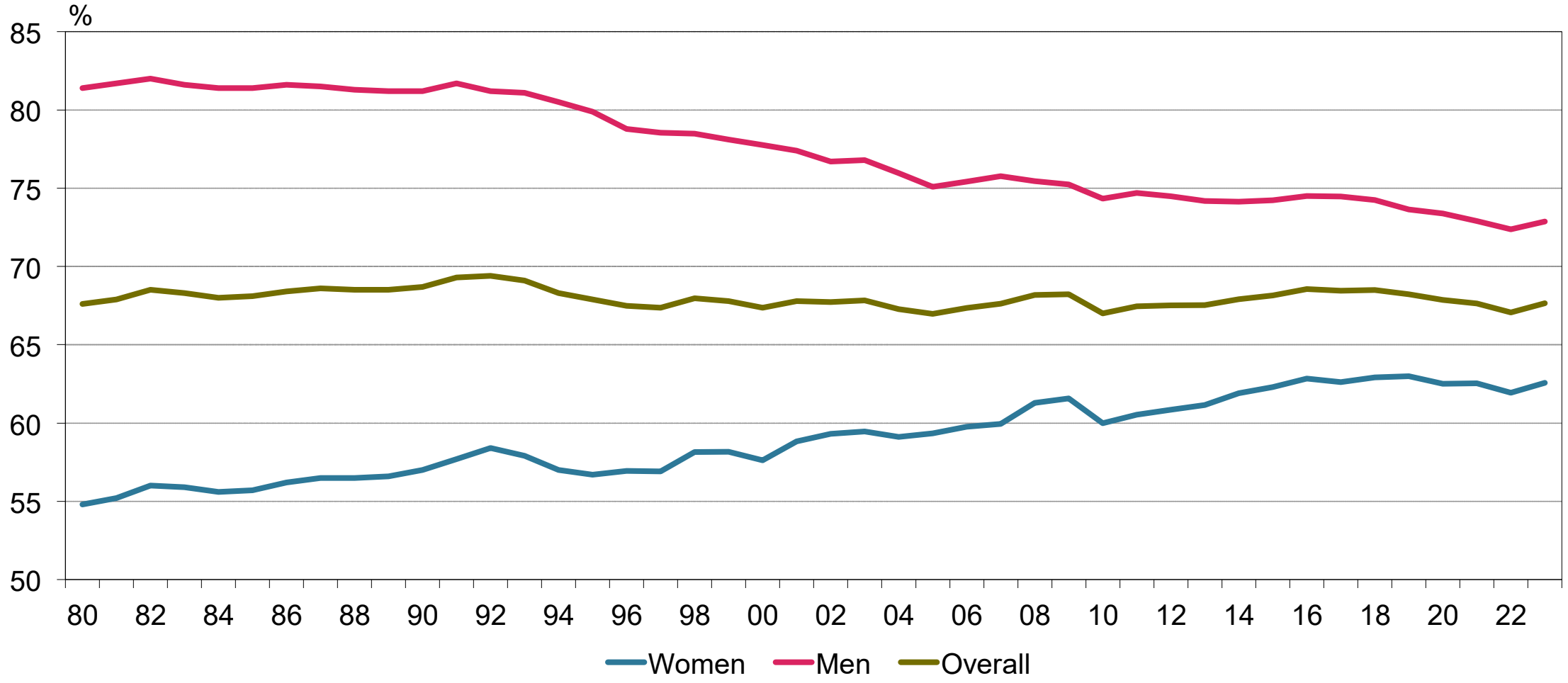
Source: SFSO, Note: in thousands of persons

	01/01/24
Not in the labour force	3,923
Unemployed	215
Employed	5,333
	9,471
Of which: Net cross-border commuters	509
Population	8,962

(1) How Is Unemployment Measured?

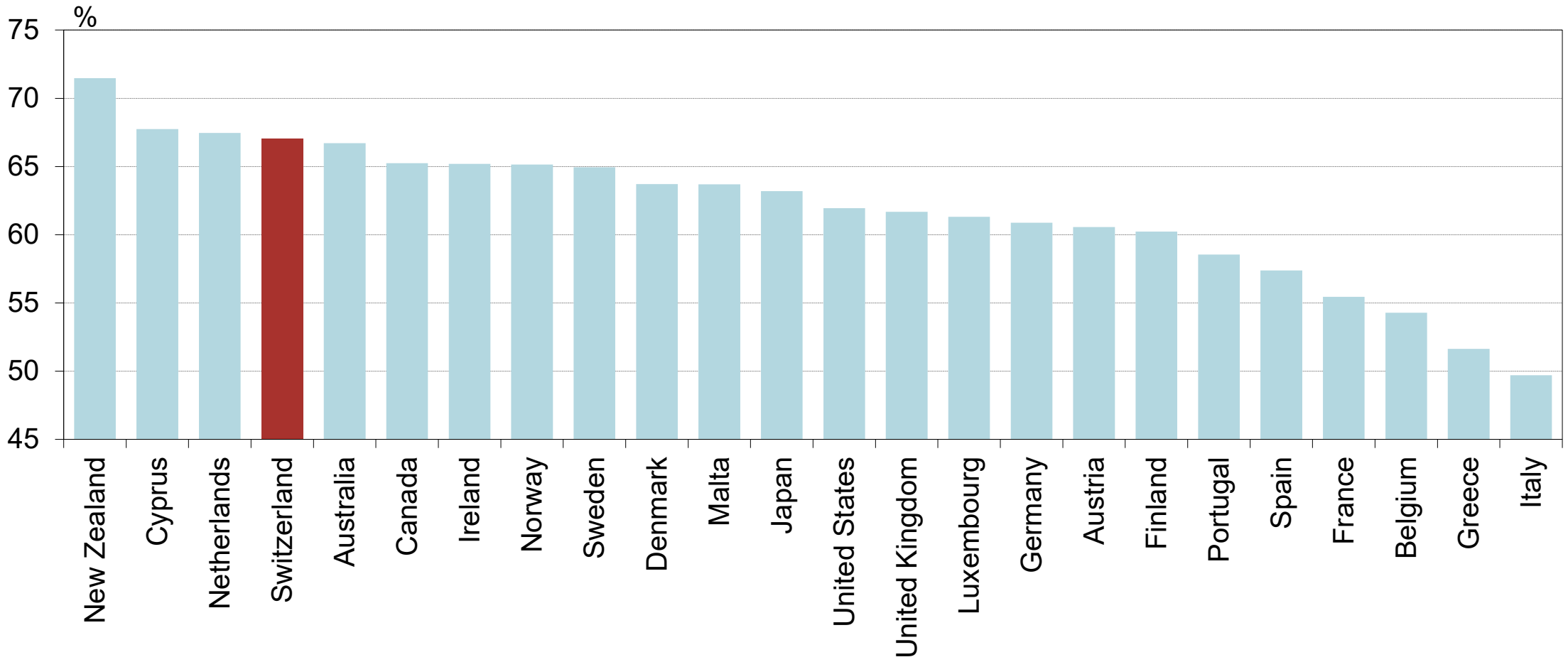
1. The ***labour force participation rate*** is the percentage of the adult population that is in the labour force
   - Labour Force Participation Rate = (Number of persons in the labour force / Adult population) x 100
     - Where any person aged 16 or older counts to the adult population (when using the standardised definition)

Labour force participation rates in Switzerland



Source: ILO (Labour force participation rate is the proportion of the population ages 15 & older that is economically active)

Labour force participation rates around the world (2024)



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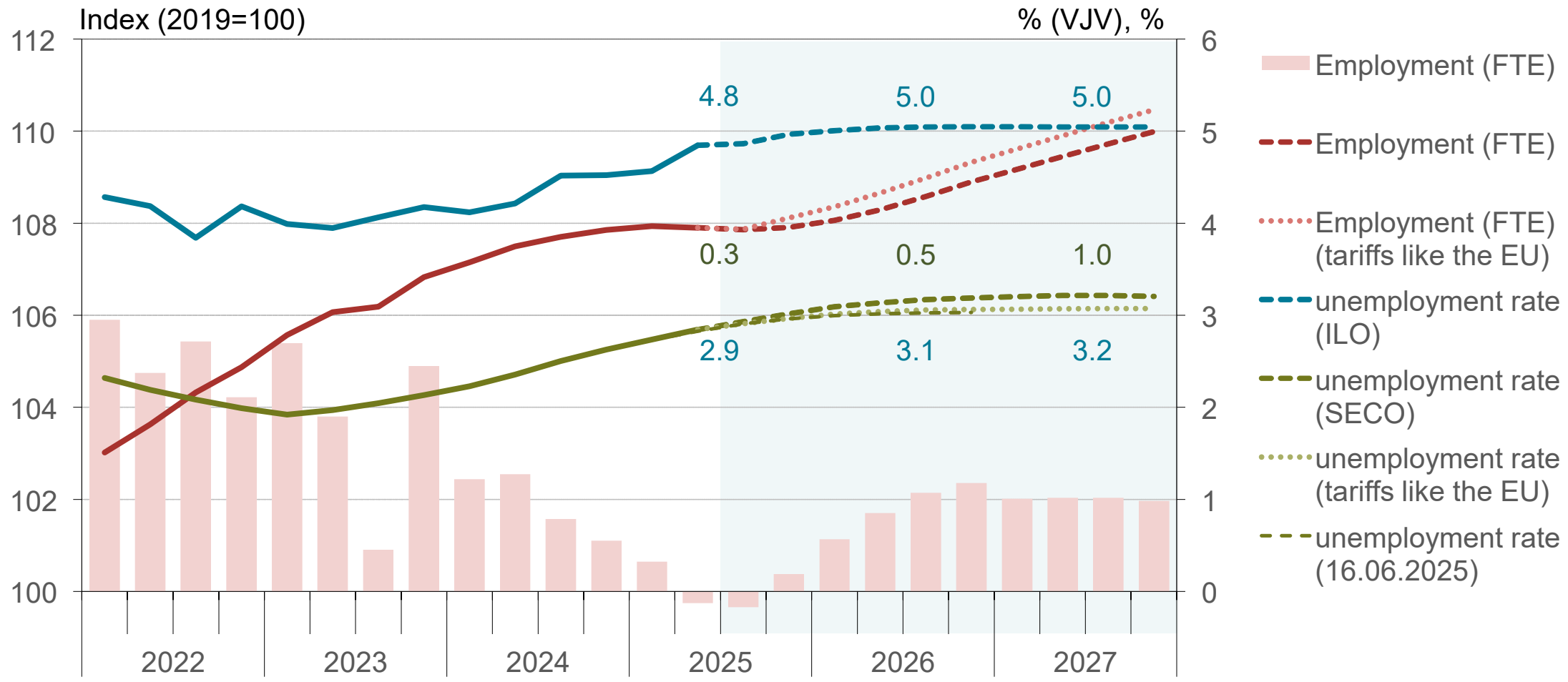
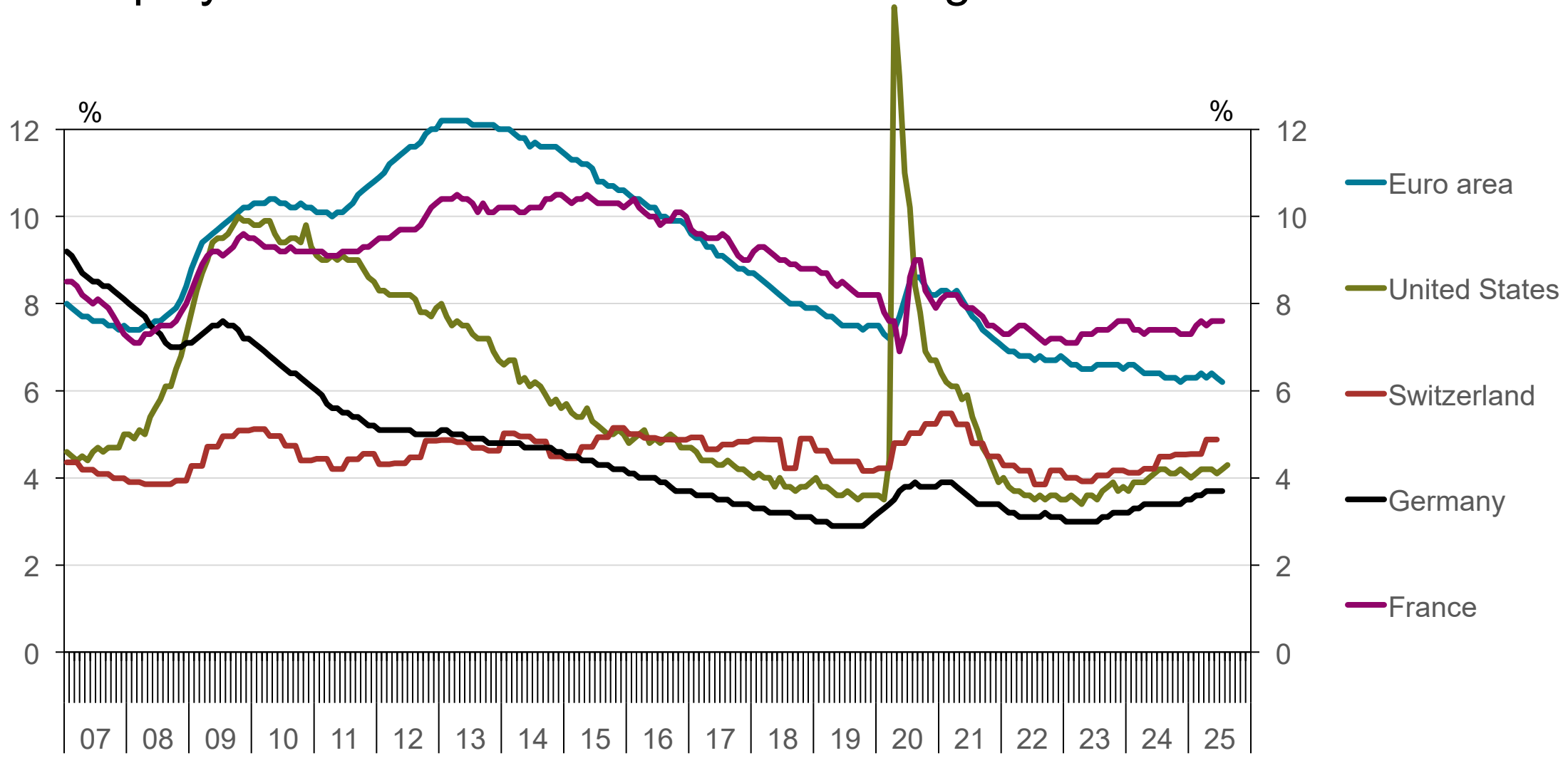
2. The ***unemployment rate*** is calculated as the percentage of the labour force that is unemployed
   - Unemployment rate = (Number of unemployed / Labour force) x 100
   - In practise measured in two ways
     - The claimant count
       - In Switzerland done by SECO (“Arbeitslosenquote”)
     - A monthly survey of households
       - In Switzerland done by the SFSO (ILO based figures) (“Erwerbslosenquote”)

Labour market developments in Switzerland



Quellen: SECO, BFS, KOF (24.09.2025)

Unemployment rate in western countries/regions



Source: OECD, Eurostat

(2) Does the Unemployment Rate Measure What We Want It To?

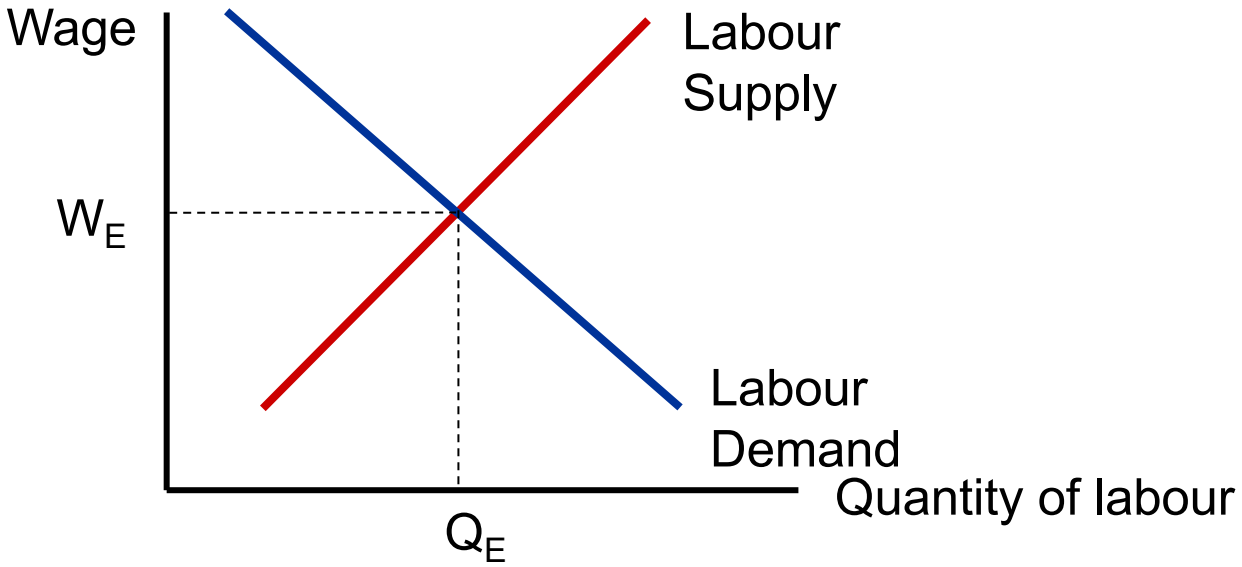
- Problem: It is difficult to distinguish between a person who is unemployed and a person who is not in the labour force
  - ***Discouraged workers***, people who actually would like to work but are discouraged and have given up looking for jobs after unsuccessful searches. Consequently, these don't show up in unemployment statistics anymore and are – with the current measures – considered not to be in the labour force
  - Other people may claim to be unemployed in order to receive financial assistance, even though they aren't looking for work or may even have (unregistered) working opportunities
- These instances may confound our measures

(3) How Long Are the Unemployed without Work?

- In judging how serious the unemployment problem is, we need to consider the duration of unemployment
  - Economists have devoted much energy to study data on unemployment duration and come up with an important, subtle and seemingly contradictory finding:
    - «Most spells of unemployment are short,
      and most unemployment observed at any given time is long term.»
  - Hence, most people who become unemployed will soon find jobs.
    Yet most of the economy's unemployment problem is attributable to relatively few workers who are jobless for long periods of time
    - ***Hysteresis*** refers to the effect is that the longer people are without work the less likely they are to be hired by firms

(4) Why Are There Always Some People Unemployed?

- In an ideal labour market, wages would adjust to balance the supply and demand for labour, ensuring that all workers would be fully employed
- In equilibrium there is no unemployment, because everybody who is willing to work at the given market wage is able to do so (market clears)
  - Everybody who is not working does not want to and will not look for work at current wage levels

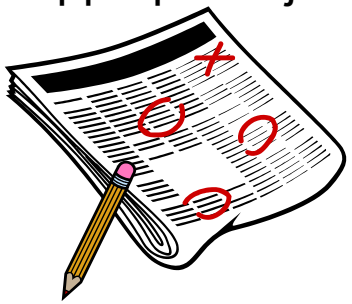


(4) Why Are There Always Some People Unemployed?

- Two different types of unemployment are responsible for long-term unemployment:
  - ***Frictional unemployment***
    - Unemployment that results from the time that it takes to match workers with jobs that are best suit their tastes and skills
    - This type is often thought to explain relatively short spells of unemployment
  - ***Structural unemployment***
    - Unemployment that results because the number of jobs available in some labour markets is insufficient to provide a job for everyone who wants one (labour supply exceeds labour demand)
    - It is often thought to explain longer spells of unemployment.
- We examine four different ways to explain unemployment in the long-run:
  - Frictional unemployment due to (A) job search
  - Structural unemployment due to (B) minimum wage laws
    (C) unions
    (D) efficiency wages

(4A) Job Search

- One reason why economies always experience some unemployment is job search.
  - ***Job search*** is the process by which workers find appropriate jobs given their tastes and skills
  - It results from the fact that it takes time for qualified individuals to be matched with appropriate jobs
  - This unemployment is different from the other types of unemployment:
    - It is not caused by a wage rate higher than equilibrium (as in the case of structural unemployment)
    - It is caused by the time spent searching for the “right” job

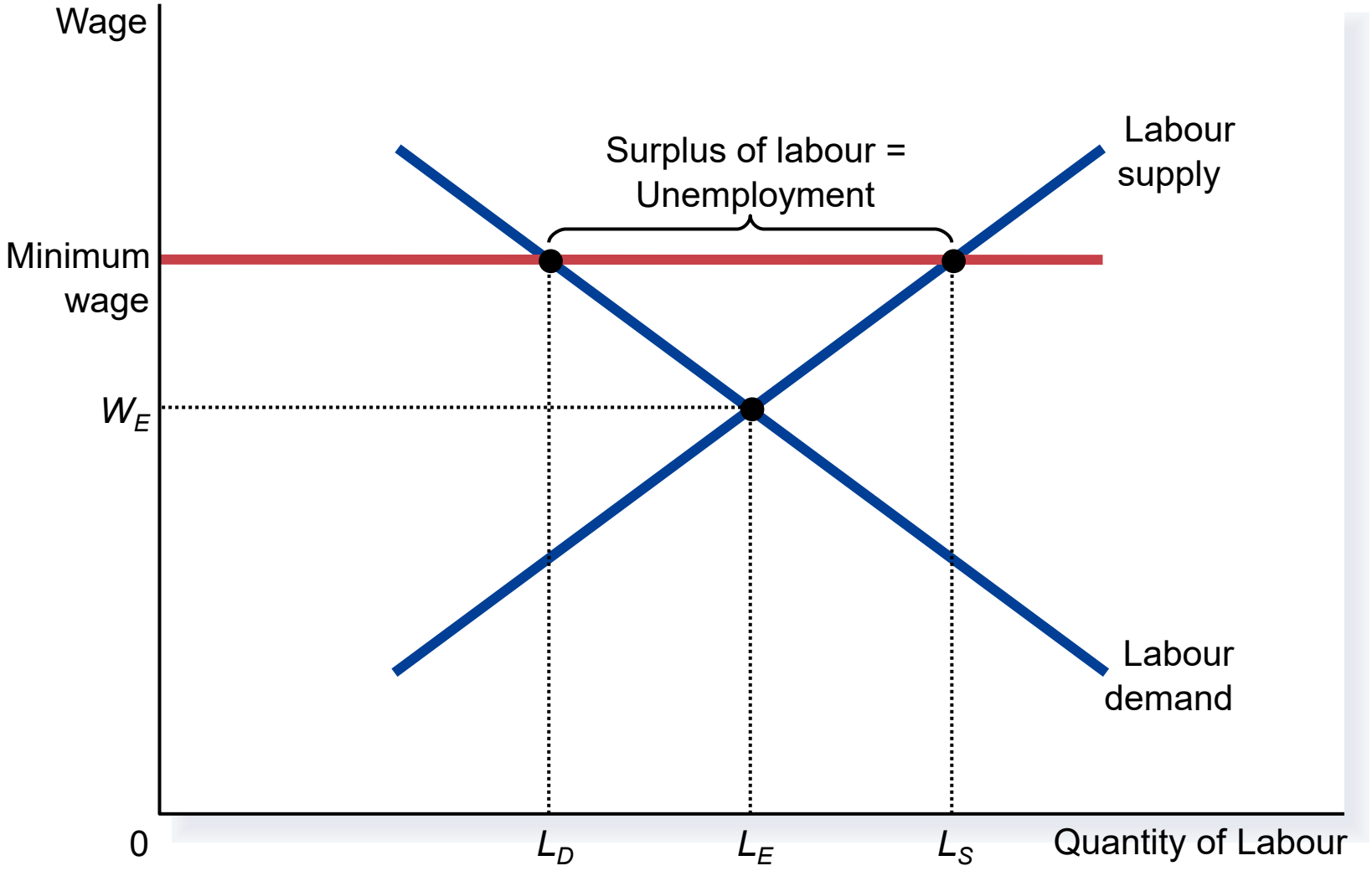


- Job search related frictional unemployment is inevitable because the economy changes continuously
  - Workers leave their jobs because they realize that the jobs are no a good match
  - If consumers start prefer Brand A over Brand B, then B lays workers of while A hirers workers
  - Technological and changes in preferences lead to sectoral shifts
    - If the oil price falls, oil extracting firms cut back their production and employment. At the same time, stimulated car sales cause car manufacturers to increase their production and employment
  - …

(4A) Job Search and Public Policy

- Government programs can affect the time it takes unemployed workers to find new jobs and firms to find the right people
- These programs include the following:
  - Government-run ***employment agencies*** give out information about job vacancies in order to match workers and jobs more quickly
  - Public ***training programs*** aim to ease the transition of workers from declining to growing industries and to help disadvantaged groups escape poverty.
  - ***Unemployment insurance*** is a government program that partially protects workers' incomes when they become unemployed.
- There are advocates and critics of these policies:
  - Advocates believe the economy to operate more efficiently by keeping labour force more fully employed and by reducing the economy's inherent inequities
  - Critics question government intervention on grounds of efficiency and condemn that it may shrink the incentives to become re-employed

(4B) Minimum-Wage Laws



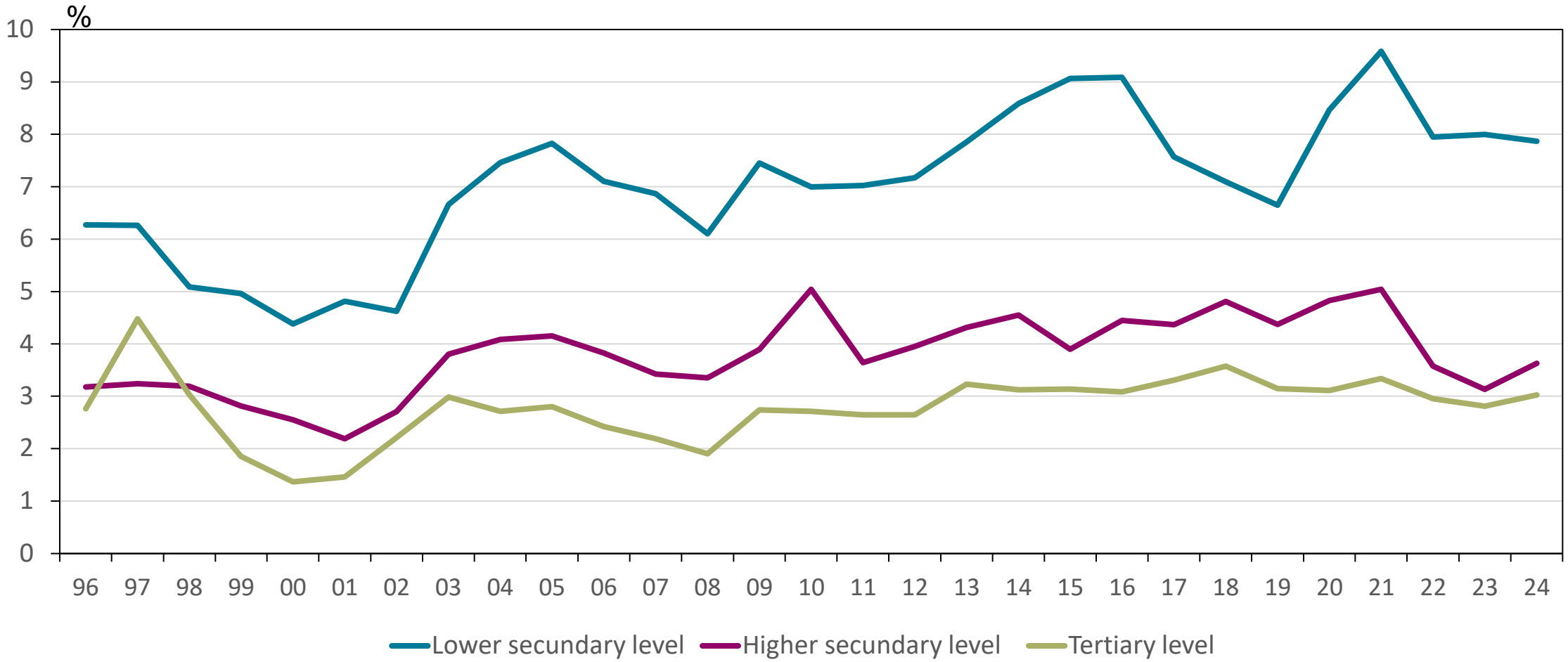
- When the minimum wage is set above the level that balances supply and demand (equilibrium level), it creates unemployment
- Social welfare benefits function like a minimum wage

Case Study: Minimum wage increase in the U.S. in 1996

Unemployment rates, before & after		
	3rd Q 1996	1st Q 1997
Teenagers	16.6%	17.0%
Single mothers	8.5%	9.1%
All workers	5.3%	5.3%

- In Sept 1996, the minimum wage was raised from $4.25 to $4.75
- Other US studies find that a 10% increase in the minimum wage increases teenage unemployment by 1 to 3%

Unemployment rate by education level in Switzerland



Source: SFSO Data represent the second quarter of each year

(4C) Unions and Collective Bargaining

- A ***union*** is a worker association that bargains with employers over wages, benefits and working conditions
- A union is a type of cartel attempting to exert its market power
  - Workers in a union act as a group when discussing their wages, benefits, and working conditions with their employers
- The process by which unions and firms agree on the terms of employment is called ***collective bargaining***
- If the union and firm do not reach agreement, the union can pressure employers to provide a higher wage level by threatening with strikes, i.e. the union organizes a withdrawal of labour from the firm
- By raising the wage level above equilibrium level (what they usually do), unions makes some workers better off and other workers worse off
  - Workers in unions (insiders) reap the benefits of collective bargaining, while workers not in the union (outsiders) bear some of the costs, e.g. unemployment. (***Insider-Outsider-Problem***)
- Empirical Evidence: Union workers earn 10 to 20 percent more than non-union workers

Are Unions Good or Bad for the Economy as a Whole?

- Economists disagree about whether unions are good or bad for the economy as a whole.
  - Critics argue that unions behave as cartels, cause the allocation of labour to be inefficient and inequitable:
    - Wages above the competitive level reduce the quantity of labour demanded and cause unemployment (inefficient)
    - Some workers benefit at the expense of other workers (inequitable)
  - Advocates of unions contend that unions are a necessary antidote to the market power of firms that hire workers. Example: Dominant employers
    - They claim that unions are important for helping firms respond efficiently to workers' concerns because unions represent workers' collective views on many job attributes and may help in finding the right mix of job attributes
- No consensus:

  Union's influence may be beneficial in some circumstances and adverse in others

(D) The Theory of Efficiency Wages

- ***Efficiency wages*** are above-equilibrium wages paid by firms in order to increase worker productivity.
- The theory of efficiency wages states that firms operate more efficiently if wages are above the equilibrium level. A firm may prefer higher than equilibrium wages for the following reasons:
  - Worker health:
    - Better paid workers eat a better diet, are healthier and are thus more productive.
  - Worker turnover:
    - A higher paid worker is less likely to look for another job. Firms save training costs, experienced worker are more productive than newly hired workers.
  - Worker quality:
    - Higher wages attract a better pool of workers to apply for jobs, because high-quality workers have higher reservation wages.
  - Worker effort:
    - Higher wages motivate workers to put forward their best effort.
- Real World Example: In 1914, Henry Ford's 5$ Workday (twice the prevailing wage of $2.34)

The Costs of Unemployment

- Costs for the individual
  - Loss of earnings
    - Increased risk of slipping into poverty
  - Self-esteem and health problems
  - Drug abuse and alcohol abuse and crime
  - Family breakdown
  - De-skilling
    - The individual loses touch with changes in work practices and technology

- Costs for society and the economy as a whole
  - The opportunity cost of unemployment
    - Lost output
  - The tax and benefits effect
    - Lower tax revenues,
      higher welfare payments
  - The Reverse Multiplier Effect:
    - People experiencing unemployment
      - Cut back their spending on luxuries
        - Causing these businesses to cut back orders, lay off workers, reduce profits
      - Switch their spending to substitute goods which may be seen as inferior goods
        - So, some firms might see demand increase e.g. budget supermarkets

Summary

- The unemployment rate is the percentage of those who would like to work but don't have jobs
- The unemployment rate is an imperfect measure of joblessness
- In the Swiss economy, most people who become unemployed find work within a short period of time
- Most unemployment observed at any given time is attributable to a few people who are unemployed for long periods of time
- One reason for unemployment is the time it takes for workers to search for jobs that best suit their tastes and skills
- A second reason why our economy always has some unemployment is minimum-wage laws
  - Minimum-wage laws raise the quantity of labour supplied and reduce the quantity demanded
- A third reason for unemployment is the market power of unions
- A fourth reason for unemployment is suggested by the theory of efficiency wages
  - High wages can improve worker health, lower worker turnover, increase worker effort, and raise worker quality